

AUTOGATE SMART PARKING SYSTEM

Official System Architecture, Operational Rules, and Knowledge Base

1. System Overview & Core Philosophy

AutoGate is an enterprise-grade, edge-to-cloud hybrid Automated Number Plate Recognition (ANPR) and Smart Parking Management System. The system replaces obsolete manual ticketing methodologies with real-time computer vision processing streams to optimize vehicle throughput, ensure ironclad security, and maintain rigorous data auditing at ingress and egress boundaries.

Core Innovation: OCR-Free High-Speed Character Extraction

Standard ANPR pipelines rely on heavy, latency-prone third-party OCR engines (e.g., Tesseract or EasyOCR), which frequently fail under unconventional license plate configurations. AutoGate bypasses this limitation by training an integrated object detection model (YOLOv8) to detect individual alphanumeric characters natively. A custom X-Y Coordinate Geometry Sorting Algorithm is then deployed to automatically parse multiple text lines, yielding near 100% string reconstruction accuracy with millisecond processing latency.

2. Official Parking Regulations & Tariffs

The AutoGate system actively enforces university and campus parking protocols based on vehicle registration status. It automatically classifies incoming scans against the central PostgreSQL datastore to determine entry authorization and tariff calculations.

USER CATEGORY	TARIFF RATE	ACCESS RESTRICTIONS	GRACE PERIOD
Registered Staff	Free / Exempt	Unrestricted access to all faculty zones	N/A
Registered Student	Free / Exempt	Designated general parking zones only	N/A
Campus Administrator	Free / Exempt	Reserved administrative parking bays	N/A
General Outsider	Rs. 50 / Hour	Public zones only. Ticket generated at entry	15 Minutes
Unregistered / Anomaly	Rs. 200 Penalty	Held at barrier. Subject to security review	Zero

3. Technical System Architecture

The solution comprises four core technical layers working synchronously:

- **Edge Inference Layer:** Captures video frames from physical RTSP network IP cameras, converts them via OpenCV streams, and runs inference through a localized model endpoint.
- **Backend API Layer (Flask):** A robust RESTful microservice architecture that exposes structured routes (such as `/api/parking/event`) to validate payloads, trigger business logic, and compute logs.
- **Persistent Storage Layer (PostgreSQL):** Manages structured schema relations across tables, primarily utilizing the `parking_logs` relations to handle entry/exit telemetry timestamps, plate tokens, and confidence matrices.
- **Frontend Monitoring Layer (ReactJS):** A live-updating client dashboard providing administrators with occupancy gauges, real-time analytics graphs, anomaly triggers, and manual override capabilities.

4. Algorithmic Processing & Filtering Logic

To operate reliably in real-world environments without crashing or spamming infrastructure databases, AutoGate implements strict filtering layers within the localized ingestion script:

Ingestion Rule Constraints

1. Character Length Filter: REJECT if `len(plate_text) < 4`
(Eliminates sporadic environmental noise and partial vehicle frame clipping)
2. Confidence Threshold Filter: REJECT if `avg_confidence < 60%`
(Ensures blurry frames or severe headlight glares do not write corrupted text values)
3. State Memory Retention: REJECT if `plate_text == last_saved_plate`
(Prevents a stationary vehicle waiting at a raised barrier from sending duplicate database inputs)

Troubleshooting Environmental Failures

If a valid vehicle approaches the gate but the system fails to extract characters, operators must ensure: (a) The physical camera lens is clean and free of condensation, (b) High-intensity headlight glare is mitigated via appropriate camera angles, and (c) The license plate utilizes alphanumeric characters matching the standard formatting length rules outlined above.

5. Frequently Asked Questions (FAQ) for Users

Q: What happens if an unregistered vehicle enters the gate?

The Flask API cross-references the plate number against the `vehicles` table. If no record exists, an alert is dispatched to the Admin Dashboard under the `anomalies` relation, triggering a real-time system warning. The desktop barrier application holds the gate closed until security verifies the operator or issues a manual override.

Q: How is the total available occupancy tracked?

The system dynamically executes a distinct count query matching un-exited entry records inside the `parking_logs` table against the maximum system constant `TOTAL_PARKING_SPOTS`. This allows the chatbot and UI to accurately broadcast open slots down to the millisecond.